Errors reveal the depth of human reasoning

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Abstract

How deeply a person deliberates is central to theories of bounded rationality, yet rarely measurable in natural behaviour. Here we show that the depth of human reasoning can be read from the structure of the mistakes people make. Using millions of skill-graded chess decisions scored against an engine at progressively deeper levels of analysis, we find that players of different skill differ little on the moves they get right but are sharply separated by the depth at which their errors are exposed—the *depth of satisficing*. In slower time controls this depth rises with skill and collapses under time pressure, vanishing in fast play. Fit without any timing information, it tracks how long players think, and it recovers the known depth of synthetic players. Incorporating it improves prediction of human moves where deep reasoning can vary. What people get wrong, more than what they get right, reveals how they think.

Keywords: bounded rationality, satisficing, depth of reasoning, decision-making, expertise, chess

Author note.

Results report held-out analyses on the full dataset; strata and the primary interaction were fixed before held-out evaluation. Evaluation is split by player throughout. T. Maharaj formerly published as T. T. Biswas.

1 Introduction

A long tradition in the decision sciences holds that people do not optimise but *satisfice*: they search through options until one crosses an acceptability threshold, then stop and choose [1, 2]. The idea recurs across the fast-and-frugal program [3], resource-rational analysis that casts cognition as the optimal use of limited computation [4–6], heuristics-and-biases accounts of judgement [7, 8], and the level-k and cognitive-hierarchy models of strategic thought [9–15]—all of which turn on one hidden quantity: how far a decision maker looks before committing. That quantity has been inferred almost exclusively from small staged experiments in artificial settings. A notable recent exception used a controlled board game to show, through behavioural modelling, that planning depth grows with expertise [16, 17]; whether such regularities govern real, repeated, high-stakes choice—outside the laboratory and at population scale—remains hard to test, because natural decisions rarely come with a ground-truth measure of how good each option was, let alone how good it *looked* at successive depths of consideration.

Chess supplies exactly this, and has long been the model system for studying expert cognition, memory, and search [18–22]. Millions of decisions by players of precisely known skill are recorded in real competition, and each can be scored against an engine far stronger than any human [23, 24]—at every depth of an iterative-deepening search. The engine's evaluation of a move *changes with depth*: a move that looks strong shallow may be refuted only deeper, and a move that looks weak at first may reveal its worth later. This depth-resolved record lets us ask, of every human mistake, *at what depth the flaw becomes visible*, and therefore how deep the player must have looked.

Our central construct is the **depth of satisficing**: the depth at which the move a player chose stops looking better than the move they should have chosen. We show that this is the dimension along which skill is most legible. Players of widely different ratings are nearly indistinguishable on decisions they get right; the information separating them is concentrated in the small fraction of moves on which they err, and specifically in the depth at which those errors surface.

We make three advances over prior chess-error and engine-depth work [25–27]. First, we replace descriptive curve-fitting with a generative model in which the player is a latent distribution over search depths, so depth of satisficing is a posterior quantity with stated uncertainty. Second, we validate that engine depth is a meaningful proxy for human reasoning depth in three independent ways—predictive, convergent, and differential—rather than assuming it. Third, we use inferred depth to build interpretable individual profiles and to predict held-out human moves, outperforming a strong state-only behavioural model precisely where theory says deep reasoning should matter, but not elsewhere; this is notable given that strong searchless models exist [28].

2 Results

Skill lives in the errors, not the successes.

Across 922,412 decisions spanning ratings 1200–2600+, the error of the played move is nearly indistinguishable across skill at shallow depth and diverges only as depth increases. Because Lichess ratings are pool-specific (a separate rating per time control), we read this *within* each control: the mean regret of the played move at full depth falls with rating in classical (from 0.059 at the weakest band to 0.028 at the strongest), rapid, and blitz, and is essentially flat in bullet ($0.070 \rightarrow 0.055$), where time precludes deep search (Fig. 1). The discriminating information is concentrated in the depth structure of mistakes, and specifically in *deep-discovery* choices: a single "swing-up" decision—a played move that looks unremarkable shallow but proves best only deeper—is several-fold more rating-predictive than a "swing-down" (trap) decision (pool-normalized single-decision $R^2 \approx 0.006$ vs. 0.003; ≈ 0.033 vs. 0.009 on the raw rating scale), with the same direction on the 2026-05 replication. The signal lies in *at what depth a move's worth appears*—the very swing-up choices on which the depth-aware model's predictive gain concentrates (below)—not in the size of the error.

Depth of satisficing rises with skill and collapses under time pressure.

Because stronger players are over-represented in slow play, this relationship must be read *within* a time control; pooling conflates skill with available clock. Within classical games the posterior depth of satisficing rises from $d \approx 10.8$ at the lowest band to $d \approx 13.7$ at the highest, with non-overlapping 95% credible intervals between the extreme bands (Spearman $\rho = +0.43$; rapid +0.53, blitz +0.43; Fig. 2a). In *bullet* the relationship vanishes ($\rho = -0.03$, flat across bands)—a built-in negative control: when no one has time to search, depth no longer separates skill, ruling out a pure rating artefact. The same logic appears within players: de-meaned depth rises across think-time quartiles ($-0.11, -0.10, -0.07, +0.27$ plies, fast to slow; Fig. 2b).

Inferred depth tracks real thinking time.

This is the construct's key non-circular test. A model fit with the clock feature removed—so timing can play no role in the estimate—nonetheless yields a depth that correlates with the actual time the player spent on the move (held-out Spearman $\rho = +0.39$; opening +0.35, middlegame +0.44, endgame +0.30; Fig. 3), strongest in the middlegame where deliberation varies most. Inferred depth tracks real thinking time without ever being told about time—convergent evidence that it reflects deliberation rather than the rating prior.

A depth-aware model predicts human moves better, only where depth should matter.

A fusion of a state-only model (Maia-3) and a search-only mixture-over-depths model improves held-out cross-entropy over Maia-3 alone by a modest 0.012 nats overall (95% CI $[0.008, 0.016]$; top-1 match +0.7 points). The interest is not the average but the *differential*, registered in advance: the gain is concentrated in classical, high-swing positions (+0.053 nats) and is absent—even slightly negative—in fast or low-swing play (Fig. 4). The swing-magnitude $\times$ time-control interaction is $+0.059$ nats (95% CI $[0.054, 0.063]$; player-clustered bootstrap), confirmed by a mixed-effects model with players as random effects ($\beta = 0.057$ nats, $p < 10^{-200}$). A state-only model cannot produce this pattern. The gain further localises by swing *direction*: within classical play it falls almost entirely on swing-up moves ($+0.048$ nats)—those whose worth appears only at depth—so the depth signal predicts precisely the choices a pattern model cannot.

An item-response reading: positions are items, swing is discrimination.

These results admit a measurement-theoretic formulation that unifies them and places people and positions on *one* scale. Treat each decision as a test item whose *difficulty* is its *critical depth*—the deepest search depth at which a misleading move still appears best—measured in plies, the same unit as the depth of satisficing; the item *response* is the ordered severity of the move played (best / inaccuracy / mistake / blunder). An explanatory graded-response model [29], with item properties entered as engine-derived covariates rather than free per-item parameters to suit the one-player-per-position design (a linear logistic test model [30]), recovers exactly the expected item-response structure: error severity falls with player ability ($\beta = -0.13$, $p < 10^{-80}$), rises with item difficulty ($\beta = +0.06$, $p < 10^{-24}$), and—decisively—ability is *more* discriminating in high-swing positions (ability $\times$ swing $\beta = -0.08$, $p < 10^{-38}$). In the language of item-response theory, *swing is item discrimination*: test information concentrates in high-swing decisions (Fig. 7a), with the graded move-quality response shifting cleanly along the ability axis (Fig. 7b)—which is precisely why skill is legible only in that small fraction of positions. The depth of satisficing thus reads as a latent *ability* on a common ply ruler shared with item difficulty (Fig. 7c)—a natural generalisation of rating systems, themselves a one-parameter item-response (Rasch) model on game *outcomes* [31–33]. At the present decision density per player, ability is reliable only in aggregate; a person-dense, pool-invariant ability estimate is left to future work.

Individual profiles separate expert styles.

Estimated per player from an *elo-free* model (so recovering rating is not circular), a four-component profile—depth of satisficing, trap-susceptibility, deep-discovery rate, and time-elasticity—recovers rating across 873 players ($\geq$ 100 decisions each) by nested cross-validation. Because Lichess ratings are pool-specific, we recover rating *within pool* (z-scored per time control, with over-the-board play as its own pool): the four-axis profile attains held-out $R^2 = 0.10$ while the depth axis *alone* has none ($R^2 \approx 0$)—a several-fold gain from the added behavioural axes, reported on the honest within-pool scale rather than the cross-pool one (which inflates recovery by conflating skill with which pool a player is drawn from). The profile resolves a known difficulty: among elite players a one-dimensional depth score is discordant with true rating on 9 of 15 pairs, because it would rank, e.g., Abdusattorov and Nepomniachtchi above Carlsen; the multi-axis profile instead attributes their strength to different axes—Nakamura is most trap-resistant yet least time-elastic (a blitz specialist who plays deep *fast*), Carlsen most converts thinking time into depth—and no longer inverts their ratings (Fig. 5).

The instrument recovers known depth.

On synthetic agents that satisfice at fixed, known depths, recovered depth increases monotonically with the planted depth (group-level Spearman = 1.0 across planted depths 4–16; the estimator also independently recovers the agents' decision sharpness, $\hat{\beta} = 7$ vs. generating $\beta = 6$; Fig. 6). The recovery is ordinal rather than exact—per-agent estimates are compressed toward the middle and noisy (mean absolute error 3.7 plies) given ~ 70 decisions per agent—but it confirms the model measures depth rather than position difficulty. Notably, recovery requires estimating depth from the moves under a flat prior; imposing the human rating-conditioned prior collapses the estimate toward its mean, which is why the rating relationship (Fig. 2) must be corroborated by the non-circular tests above.

Pre-registered out-of-sample replication.

We pre-registered [34] the hypotheses, pipeline, and tests, then ran the identical pipeline once on an untouched month (Lichess 2026-05; 179,360 decisions). The two non-circular results replicated: clock-free depth tracked thinking time, strongest in the middlegame (held-out $\rho = +0.30$ overall; opening $+0.32$, middlegame $+0.35$, endgame $+0.20$), and the registered swing $\times$ time-control interaction held ($+0.049$ nats, 95% CI $[0.039, 0.056]$; mixed-effects $p < 10^{-36}$). Depth of satisficing rose with rating in classical ($\rho = +0.47$) and rapid ($+0.54$)—both stronger than in the primary month—and was null in bullet, but it did *not* hold in blitz ($\rho \approx 0$; inferred depth flat near 11.4 plies across the entire rating range, at coverage equal to rapid). The rating–depth relationship is thus confined to the slow controls where depth can actually be deployed, and our pre-registered blitz prediction was too strong. The single-decision rating-information result replicated nearly identically on both months but *opposite* to its pre-registered sign—swing-*up* (deep-discovery), not swing-down (trap), decisions carry the information; the pre-registration mis-stated the sign (a swing-label convention corrected before analysis), and we report the corrected, internally consistent result (cf. the swing-up localisation of the predictive gain above). By the pre-registered criterion—all three primary hypotheses holding—the replication is *not* declared a blanket success, because the blitz arm of the rating–depth relationship did not hold; the non-circular core (depth $\leftrightarrow$ thinking time; the registered predictive interaction) and the construct's central claims replicated.

3 Discussion

The errors people make are usually treated as noise. We find the opposite: where success is widely shared, error carries the informative variation, and its fine structure—the depth at which a flaw becomes visible—is a measurable signature of how a person reasons. Depth of satisficing scales with expertise, contracts under time pressure, and powers prediction exactly in the slow regime where depth can vary. This supports the view that people commit to the best option found at the plateau of search they reach rather than integrating across the whole trajectory, consistent with level-k choices being largely independent of higher-order belief tails [15]. Where planning depth has been measured before it has relied on controlled laboratory games and explicit behavioural search models [16, 17]; here the same construct is read non-invasively from natural, high-stakes play at population scale, against a depth-resolved value standard. The mechanism it implies—committing to the best option found at the search horizon a player reaches—links satisficing to model-based control [35], decision-tree pruning [36], and bounded evidence accumulation [37], and the individual profiles echo evidence that expert flexibility itself varies [38]. In principle the approach extends to any domain with skill-graded decisions and a depth-resolved value standard—a depth-resolved value network in Go, or staged value estimates in other tasks—which we leave as the natural next test of generality. A constructive corollary: an agent that estimates a person's depth of satisficing could tailor problems just beyond it. *Limitations:* engine search depth is a *validated proxy* for human selective search—supported here by convergence with thinking time (Fig. 3) and by recovery of planted depth (Fig. 6), but not isomorphic to it; the depth-of-satisficing posterior leans partly on the rating prior, which is why we rely on those two non-circular tests; the aggregate predictive gain is small and the claim rests on its registered differential; and chess is adversarial and highly trained, so cross-domain generality remains to be demonstrated.

4 Methods

Data.

We use 922,412 decisions (16,328 players): online games from the Lichess open database (official parquet export, month `2025-09`), balanced-sampled across rating bands (200-point bins, 1200–2600+) and time classes (bullet/blitz = fast, rapid/classical = slow), with per-move clock times retained; and an elite over-the-board stratum of 371,279 decisions from 1,261 titled players (broadcast tournament games), which supports the individual-profile case study. Synthetic agents (below) provide ground truth. All decisions are filtered to plies 9–120 to skip opening-book play.

Engine analysis and value scaling.

Positions are analysed with Stockfish 18 in Multi-PV mode, recording each candidate move m_i's evaluation $v_{i,d}$ at every depth $d = 1 \dots D$ (D=21, K=9). The candidate set is the union of top-K across all depths so trap moves keep a full trajectory. Because consecutive plies of each game are analysed, the value of a played move lying outside the top-K is read off from the position it leads to—the next sampled decision—rather than re-searched separately: its win probability at depth d equals one minus the successor's best win probability at depth $d-1$ (a half-move shallower, matching the other candidates' negamax depth). On the rare occasions when no analysed successor exists (game end or the final sampled ply) the played move is assigned the weakest retained candidate's win probability at each depth. With `UCI_ShowWDL` the engine reports win/draw/loss counts (w, dr, ℓ) in per-mille, so the win probability $P_{i,d} = (w + \frac{1}{2}\,\mathrm{dr})/1000$ is read off per move per depth—no logistic fit. Depth-d regret is

$$\delta_{i,d} = \max_j P_{j,d} - P_{i,d} \;\geq\; 0. \tag{1}$$

Because δ is a difference against the best move *at each depth*, it is invariant to anything that shifts all moves together (position difficulty, evaluation scale), isolating the choice problem. Searches run to depth D or a 2×10^8-node cap, whichever is first, a safety bound on the rare quiet endgames whose multi-PV search does not converge; 0.09% of positions hit the cap and stop short of D, and are handled by masking the unreached depths in the likelihood (their reached depth is recorded) rather than imputing.

State-tower features.

The human-pattern prior $q_i(p,c)$ is the Maia-3 policy at the side-to-move's rating, read from the policy head as a full distribution over legal moves.

Latent-depth choice model.

A player who has effectively searched to depth d chooses by a soft-max over depth-d regret fused with the pattern prior as a product of experts,

$$P(\text{move}{=}i \mid d,p,c) = \frac{\exp(-\beta\,\delta_{i,d} + \alpha\,\log q_i)}{\sum_j \exp(-\beta\,\delta_{j,d} + \alpha\,\log q_j)}. \tag{2}$$

Effective depth is latent with distribution $\pi_d(p,c) = P(\text{searched to } d \mid p,c)$, emitted by a network over a coarse depth grid conditioned on context c (rating, time class, clock, move number, total swing). The move likelihood marginalises depth,

$$P(\text{move}{=}i \mid p,c) = \sum_d \pi_d(p,c)\,P(\text{move}{=}i \mid d,p,c). \tag{3}$$

Parameters are fit by minimising the negative log-likelihood of the played move y, $\mathcal{L} = -\log\sum_d \pi_d P(y \mid d)$, with an entropy penalty on π for identifiability and β tied globally; data are split by player.

Depth of satisficing.

The posterior over effective depth given the observed move,

$$r_d = P(d \mid y,p,c) = \frac{\pi_d\,P(y \mid d)}{\sum_{d'} \pi_{d'}\,P(y \mid d')}, \qquad \hat{d} = \sum_d d\,r_d, \tag{4}$$

gives a per-decision depth with uncertainty. Lichess ratings are *pool-specific* (a separate Glicko-2 rating per time control, not comparable across controls); we therefore never pool ratings across controls—every rating analysis is run *within* a time control (pooling would also confound skill with available clock). Per-band estimates use player-clustered bootstrap [39] 95% confidence intervals (resampling players, 1000 replicates).

Relation to the curve-intersection estimator.

The construct originates in a population-level heuristic [26]: averaging over many decisions, the mean error of the played move and of the engine's own move are plotted against analysis depth and the depth of satisficing is read off their crossing point.

That estimator returns a single depth per cohort, carries no per-decision uncertainty, presupposes one time control, and—because the crossing is read from a polynomial fit—is sensitive to the fit, cohort depths shifting by ≈ 1 ply under reasonable fitting choices. We retain the same underlying signal (a played move whose error *rises* as the engine looks deeper is exactly the move *swing* of Biswas and Regan[26]) but replace the crossing with the generative posterior r_d above: a per-decision depth with credible intervals that supports the within-time-control and within-player contrasts used throughout and is identifiable against synthetic ground truth (Fig. 6). The population curve is stable only because it averages thousands of *weak* per-move signals—a single move barely constrains d—which the crossing view conceals and our identifiability test makes explicit: recovered depth is ordinal and compressed toward the prior, which is why the convergent (Fig. 3) and identifiability (Fig. 6) tests, not the posterior's absolute scale, carry the claims. The two views agree— stronger players satisfice deeper and err less—now established *within* time control, at far larger scale, with win-probability (WDL) engine evaluation, a human-pattern prior, and a pre-registered out-of-sample replication.

Item-response formulation.

Each decision is treated as an item. Item difficulty b_j is the deepest grid depth at which the apparent-best move differs from the full-depth best move (critical depth, in plies); the ordered response is the played move's full-depth regret binned as best (≤ 0.02), inaccuracy (≤ 0.05), mistake (≤ 0.10), or blunder (> 0.10) in win-probability units. We fit an explanatory proportional-odds graded-response model [29, 30] with predictors ability (within-pool rating z), b_j, position swing, and an ability $\times$ swing term; item properties enter as covariates rather than free per-item parameters because the natural-play design is sparse (most positions occur once). The model is fit on a random subsample for the MLE and the response curves computed on all decisions. Per-player ability has low split-half reliability at this decision density, so we report item-level and group-level (rating-band) effects and defer person-dense, pool-invariant ability estimation.

Profiles and rating recovery.

Per player (≥ 100 decisions): mean $\hat{d}$, trap-susceptibility (rate of played swing-down moves weighted by their shallow attractiveness), deep-discovery rate (rate of played swing-up moves), and time-elasticity (within-player slope of $\hat{d}$ vs. log think-time). To keep rating recovery non-circular, $\hat{d}$ for the profile is taken from a model fit with the rating feature removed. Rating is predicted from the standardised profile by ridge regression under nested 5-fold cross-validation (inner fold selects the penalty). The champion comparison reports the standardised components separately.

Model comparison and the differential test.

State-only (Maia-3, $\beta=0$), search-only ($\alpha=0$), and fusion predictors are fit under one player split and compared on held-out cross-entropy and top-1 match. The pre-registered hypothesis is an interaction—the fusion's gain over state-only is larger for high-swing than low-swing positions and for classical than blitz—tested with a player-clustered bootstrap and a linear mixed-effects model with players as random effects (positions are not crossed at this scale).

Convergent and recovery validation.

Convergent: refit the model with the clock feature zeroed, then correlate per-decision $\hat{d}$ with observed time-on-move (held out, by game phase). Recovery: agents play a softmax (β_{gen}=6) over fixed-depth d_{plant} win-probabilities; their games pass through the same pipeline, and a per-agent flat-prior posterior over depth (on the search likelihood, with the population sharpness β fit) is compared to d_{plant}.

Reproducibility.

Stockfish 18 (net `nn-71d6d32cb962.nnue`), Threads 1, Hash 256 MB, D=21, K=9, 2×10^8-node cap (positions analysed independently across many single-thread processes); Maia-3 `maia3-79m`; depth grid $\{2, 4, \ldots, 22\}$; Lichess month `2025-09`; sampling seed 17; train/test split by player. Strata and the primary interaction were fixed before held-out evaluation; we report results as confirmatory-on-held-out and label any post-hoc choices as exploratory.

Data availability.

All game records derive from the public Lichess open database (https://database.lichess.org; months `2025-09` for the primary analysis and `2026-05` for the replication) and from openly licensed Lichess broadcast exports of over-the-board play. The derived analysis datasets—per-position engine depth trajectories, Maia-3 policies, and the model tensors used for all results—are deposited at the Open Science Framework (OSF), [insert OSF data DOI].

Code availability.

The analysis pipeline (engine driver, dataset builder, model, the confirmatory replication script, and all figure code) and configuration files are released under an open licence at [insert code repository URL]; the pre-registration of the out-of-sample replication is archived in the same OSF project, [insert OSF registration DOI]. The engine (Stockfish 18) and Maia-3 (`maia3-79m`) are third-party open-source software.

Author contributions.

T.M. conceived the study, designed and implemented the latent-depth model and the full analysis pipeline, ran the experiments, pre-registered and ran the replication, and drafted the manuscript. K.W.R. contributed the intrinsic-performance-rating and move-swing framework underlying the construct, advised on chess-cognition methodology and on the validation design, and revised the manuscript. Both authors approved the final version.

Competing interests.

The authors declare no competing interests.

Ethics.

The study uses only publicly released, openly licensed game records of voluntary online and broadcast play; it involves no intervention and no private or personally identifying data beyond the public usernames/player names attached to those records, and so did not require ethical approval.

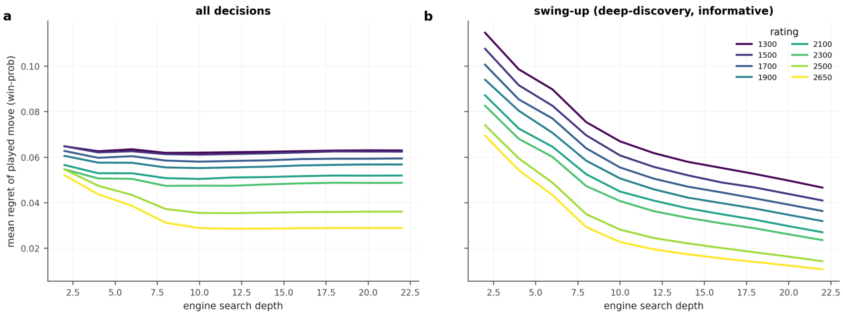

Fig. 1 Played-move regret across engine search depth, by rating band, for all decisions (left) and swing-up (deep-discovery) decisions (right), where the rating signal concentrates. Skill separates only at depth.

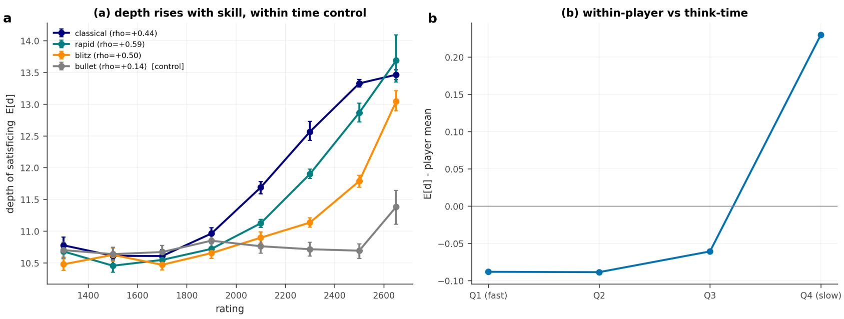

Fig. 2 (a) Depth of satisficing vs. rating, within each time control (bullet is the flat negative control); player-clustered 95% CIs. (b) Within-player depth by think-time quartile.

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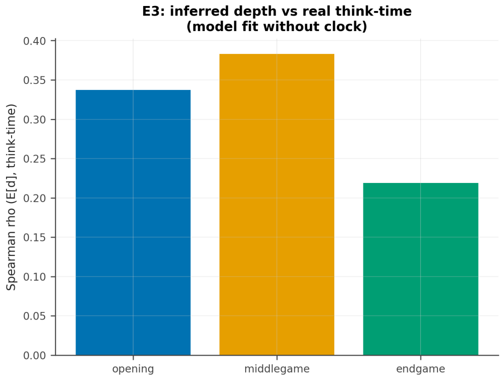

Fig. 3 Convergent validity: clock-free inferred depth vs. observed time-on-move, by phase.

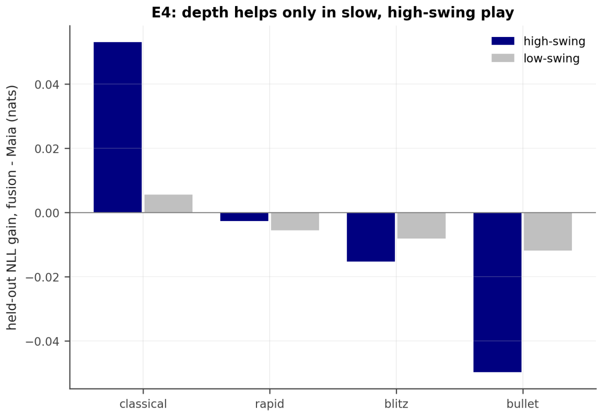

Fig. 4 Held-out prediction gain of fusion over Maia-3, by time control × swing magnitude. Positive only in classical, high-swing play.

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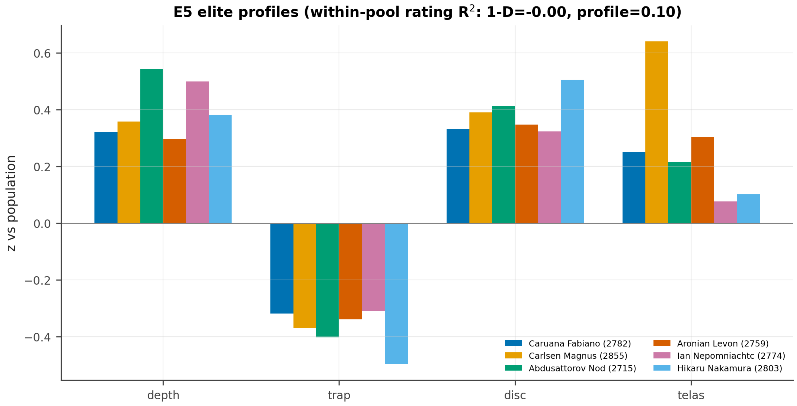

Fig. 5 Elite four-axis profiles (z-scored); a one-dimensional depth score misranks them, while the full profile recovers *within-pool* rating ($R^2 = 0.10$, nested CV; depth alone $R^2 \approx 0$).

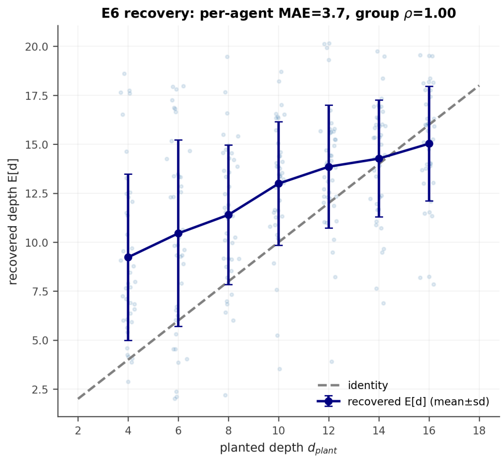

Fig. 6 Synthetic-agent recovery: recovered vs. planted depth (ordinal; group Spearman = 1.0).

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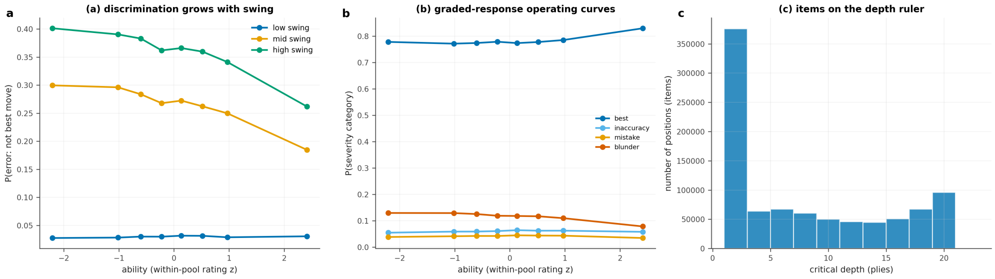

Fig. 7 Item-response view of the construct. (a) Probability of an error (not the best move) versus player ability, by position swing tercile: ability discriminates more steeply in high-swing positions—swing acts as item discrimination. (b) Graded-response operating curves: the probability of each move-quality category (best / inaccuracy / mistake / blunder) as a function of ability—P(best) rises and P(blunder) falls cleanly with skill. (c) Positions placed on the depth (ply) ruler by critical depth—the same scale on which the depth of satisficing measures person ability.

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